환경과학개론 (EVSE101)

과목 소개

2025년 2학기 후반부

1 분반: 월 수 11:00 – 12:15
2 분반: 월 수 12:30 – 13:45
청암학술정보관 멀티미디어 교육실 501호

강의 소개

9 - 16주 강의 담당 : 송우철 교수

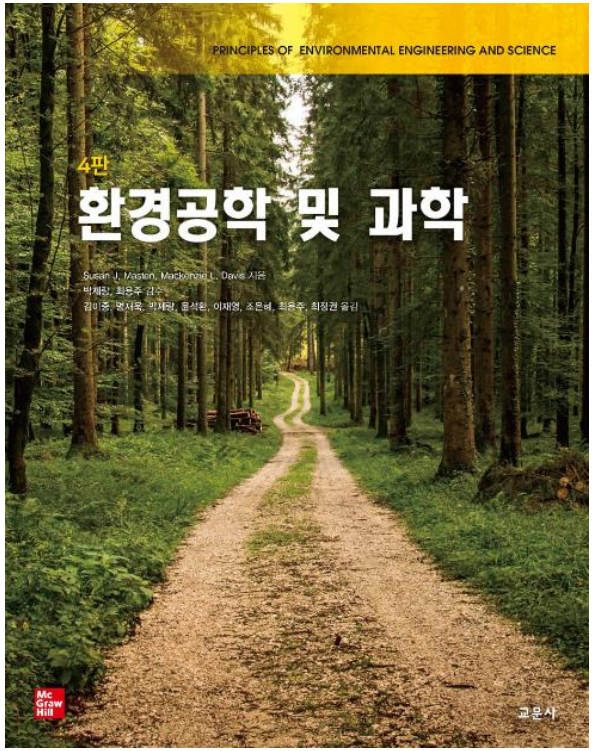
279-2281, 환경공학동 212호, woochulsong@postech.ac.kr

강의실 : 청암학술정보관 멀티미디어 교육실 501호

성적: 40% 시험 + 10% 참여 (출석 및 수업 참여)

수업 중 핸드폰 사용 금지!!

주교재



강의 목표

- "환경-에너지-기후변화"를 이해하기 위한 과학적 기초를 기반으로, 지구환경변화 및 환경문제에 대한 최근 이슈들을 소개하며, 이를 해결하기 위한 **다학제적 접근방법** 등을 학습.

- 수질 및 대기오염의 특성 및 정화기술의 원리에 대해 학습

- 환경-에너지 넥서스 기술의 원리에 대해 학습

강의 목차

- ❑환경 오염물의 순환
- ❑지표수 및 지하수의 순환 (물의 흐름)
- ❑수질 오염 및 수처리 기술
  - 수질오염의 이해 및 정화의 원리 (물 공급 & 하폐수 처리)
  - 물리화학적 수처리
  - 생물화학적 수처리
- ❑환경-에너지 넥서스 기술 (탄소 중립, 지속 가능 기술)

공학이란?

“Scientist discover the world that exists.

Engineers create the world that never was”

- Theodore von Karman



What is Engineering?

Engineering is the profession in which a knowledge of the mathematical and natural sciences gained by study, experience, and practice is applied with judgement to develop ways to utilize economically the materials and forces of nature for the benefit of mankind

- The Engineer’s Council for Professional Development (ECPD)

환경공학이란?

What is Environmental Engineering?

Environmental Engineering is manifest by sound engineering thought and practice in the solution of problems of environmental sanitation, notably in the provision of safe, palatable, and ample public w 물공급(정수) *es; The proper disposal of or recycle of w* 하폐수처리 *r and s* 폐기물관리 *s; the adequate drainage of urban and rural areas for proper sa* 위생 *ion; and the control of wa* 토양 및 대기오염 정화 *spheric pollution, and the social and environmental impact of these solutions.*

Furthermore, it is concerned with engineering problems in the field of public health, such as control of arthrop 인류 기인 질병 *eases, the elimination of industri* 산업 폐기물 *azards, and the provision of adequate sanitation in urban, rural, and recreational areas, and the effect of technological advances on the environment*

- The American Society of Civil Engineers (ASCE)

환경 공학의 시작 : 위생 관리



Sanitation refers to public health conditions related to clean drir 정수 vater and adequate treatment and disposal 배설물/오수 처리 and sewage.
Preventing human contact with feces is part of sanitation, as is hand washing with soap.
Sanitation systems aim to protect human health by providing a clean environment that will stop the transmission of disease, especially through the fecal–oral route.
For example, diarrhea, a main cause of malnutrition and stunted growth in children, can be reduced through sanitation.

Sanitation in Modern Civilization

BMJ readers choose sanitation as greatest medical advance since 1840

Annabel Ferriman BMJ

More than 11 300 readers of the *BMJ* chose the introduction of clean water and sewage disposal—"the sanitary revolution"—as the most important medical milestone since 1840, when the *BMJ* was first published. Readers were given 10 days to vote on a shortlist of 15 milestones, and sanitation topped the poll, followed closely by the discovery of antibiotics and the

pioneered the introduction of piped water to people's homes and sewers rinsed by water, attracted 15.8% of the votes, while antibiotics took 15%, and anaesthesia took 14%. The next two most popular were the introduction of vaccines, with 12%, and the discovery of the structure of DNA (9%).

A total of 11 341 people voted on the shortlist, which was chosen by a panel of experts from a list nominated

of the general public, and one in seven were students. Another tenth were academic researchers. Almost two fifths of the voters were from the United Kingdom, and a fifth were from the United States.

Johan Mackenbach, professor of public health at Erasmus MC Medical Center, Rotterdam, who championed the cause of sanitation, said, "I'm delighted that sanitation is recognised by so many people as such

portant milestone. The

ral lesson which still

s is that passive protection

against health hazards is often the best way to improve population health.

"The original champions of the sanitary revolution were John Snow, who showed that cholera was spread by water, and Edwin Chadwick, who came up with the idea of sewage disposal and piping water into homes.

"Inadequate sanitation is still a major problem in the developing world."

The *Medical Milestones* supplement is distributed with this week's *BMJ*.

111

More than 11 300 readers of the *BMJ* chose the introduction of clean water and sewage disposal—"the sanitary revolution"—as the most important medical milestone since 1840, when the *BMJ*





BMJ : 국제 의학 저널

대표적 공중 보건 및 위생 문제 사례

콜레라 (19 세기 초~)



19세기초, 인도, 아시아, 유럽, 북미 전역에서 팬데믹

1854년, 런던 시내에서 수백명 사망



John Snow 박사

상하수도 개혁



대표적 공중 보건 및 위생 문제 사례

포항 형산강 수은/중금속 오염
2016년 ~

형산강 '수은' 검출, 기준치 1만 3천 배...원인은?

입력 2017-07-29 06:47 | 수정 2017-07-29 07:21



www.youtube.com › watch

형산강 및 구무천 중금속 안정화제 시범 살포 - YouTube



포항시는 **형산강** 및 구무천 오염 의심구역에 대한 퇴적물 내 **중금속** 용출과 유동성을 억제하기 위해 **중금속** 안정화제 살포 작업을 시행했습니다.

YouTube · 포항시 · POHANG CITY · Apr 30, 2018

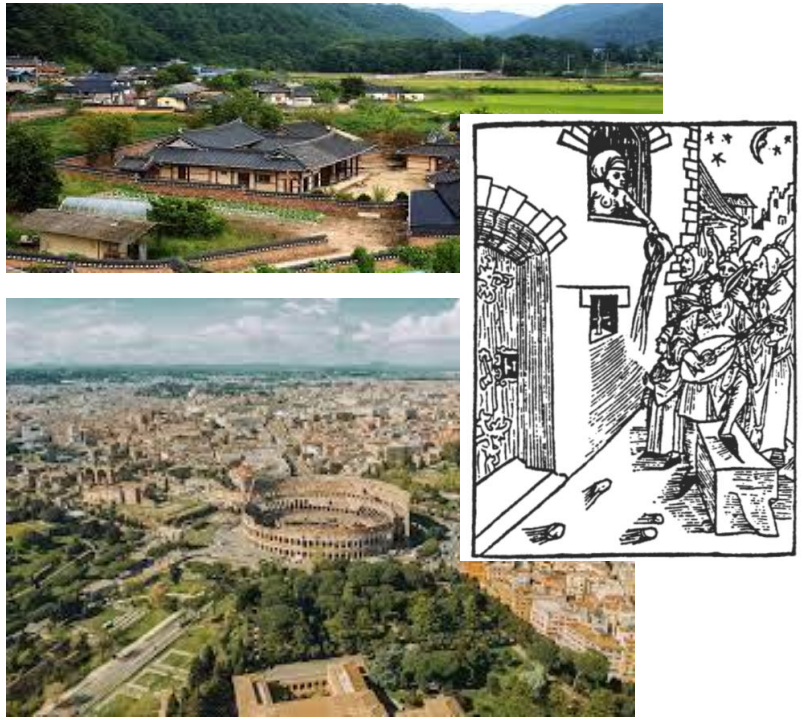
포항시, 형산강 생태복원사업 추진 속도 '박차' | 게시판 2022. 3. 23. 15:28

- 국가연계 형산강 중금속 퇴적물 정밀조사 중
- 사고발생 시 유해물질 형산강 직유입 차단을 위한 완충저류시설 설치사업 추진
- 구무천·공단천 오염퇴적물 준설 및 생태복원사업 추진
- 우·오수 분리로 하천오염 방지를 위한 하수관로 정비사업 추진



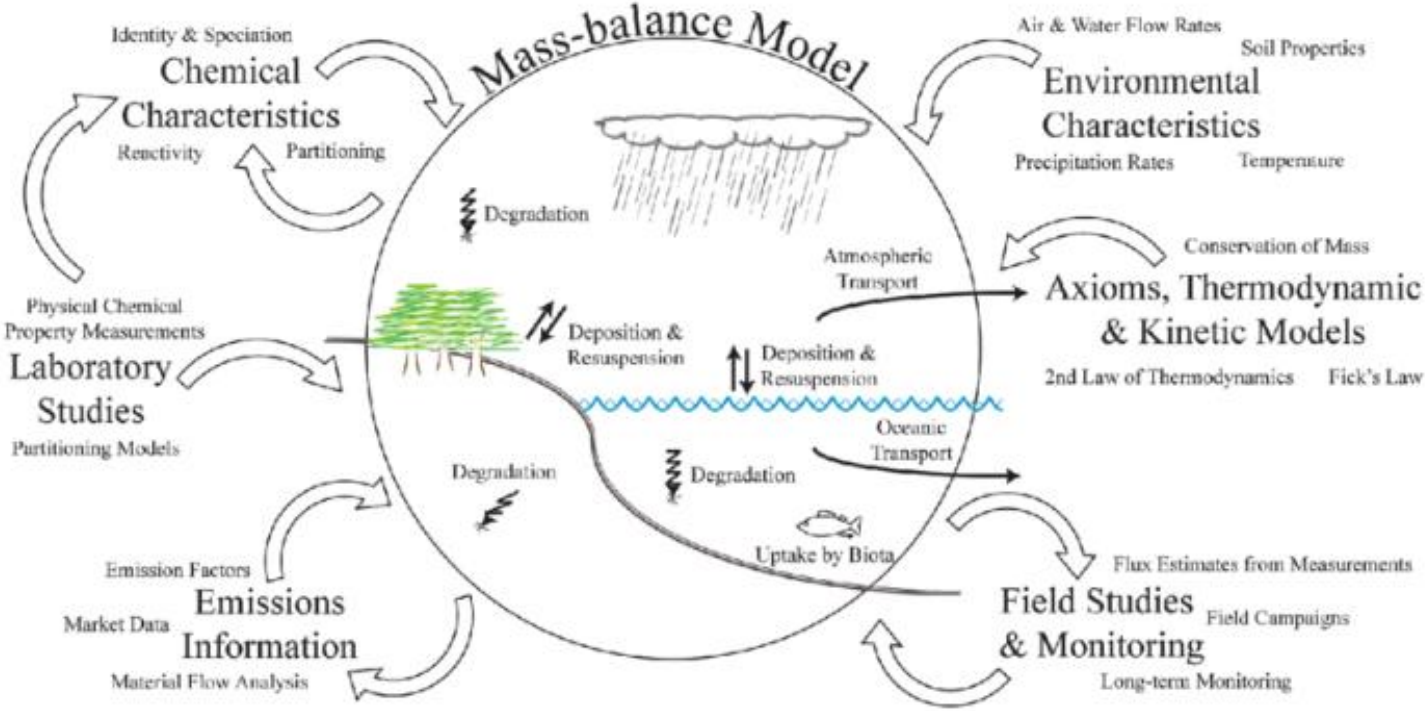
환경 오염원의 순환

기존 환경 오염원 추적 및
위생 관리 대상



마을, 국가

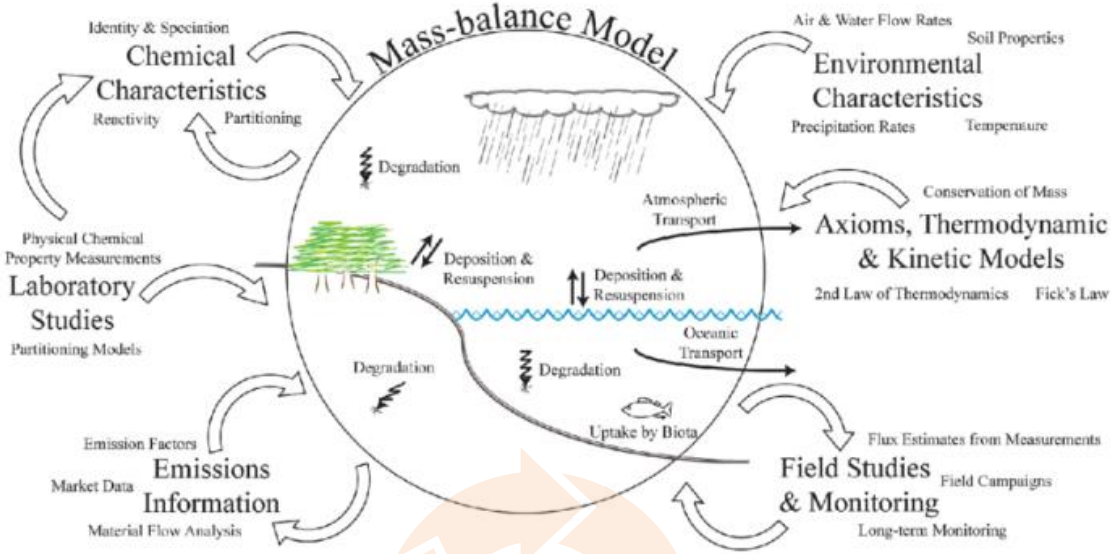
최근 환경 오염원 추적 및
위생 관리 대상

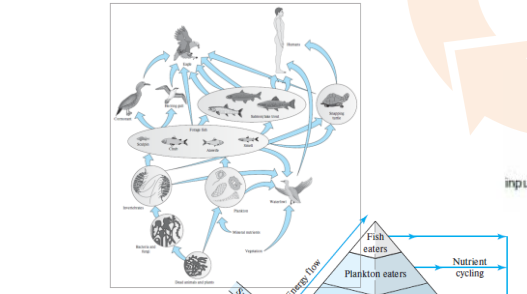


전 세계

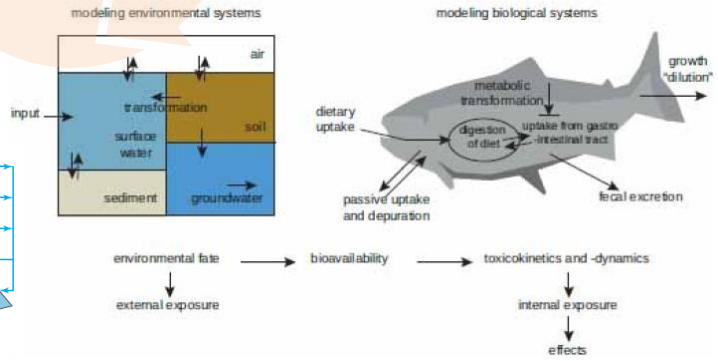
환경 오염원의 순환

오염 물질 순환





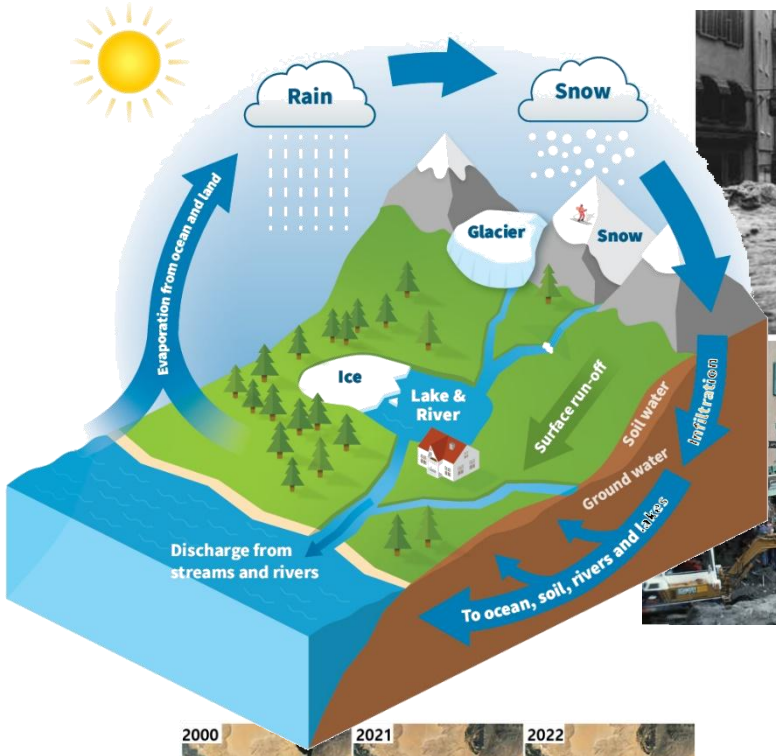
푸드 웹



- **상전이 (transfer)**
- **물질전달 (transport)**
- **화학 반응 (transformation)**

- 위해성 인지, 평가 및 관리
  - 독성평가
  - **노출 평가**
  - 위해도 결정

지표수 및 지하수의 순환 (물의 흐름)

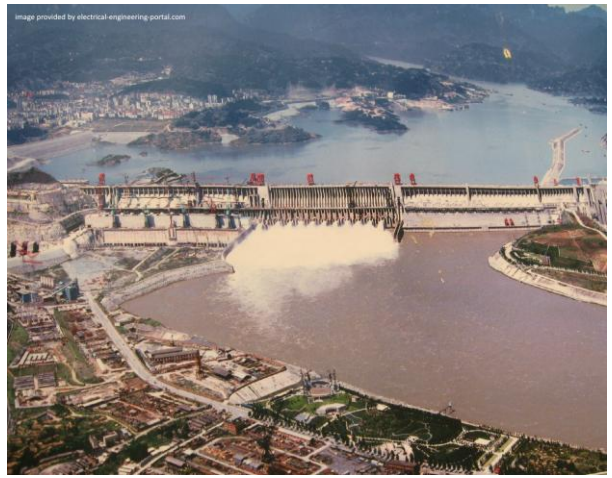












물 순환의 이해 및 수자원 관리

물이 얼마나 필요할까?

사용 가능한 물의 양이 얼마인가? (강우, 강수 등)

지구/대륙/나라 스케일에서의 물질 수지

수질 오염 및 수처리 기술

수질오염물질과 그 오염원



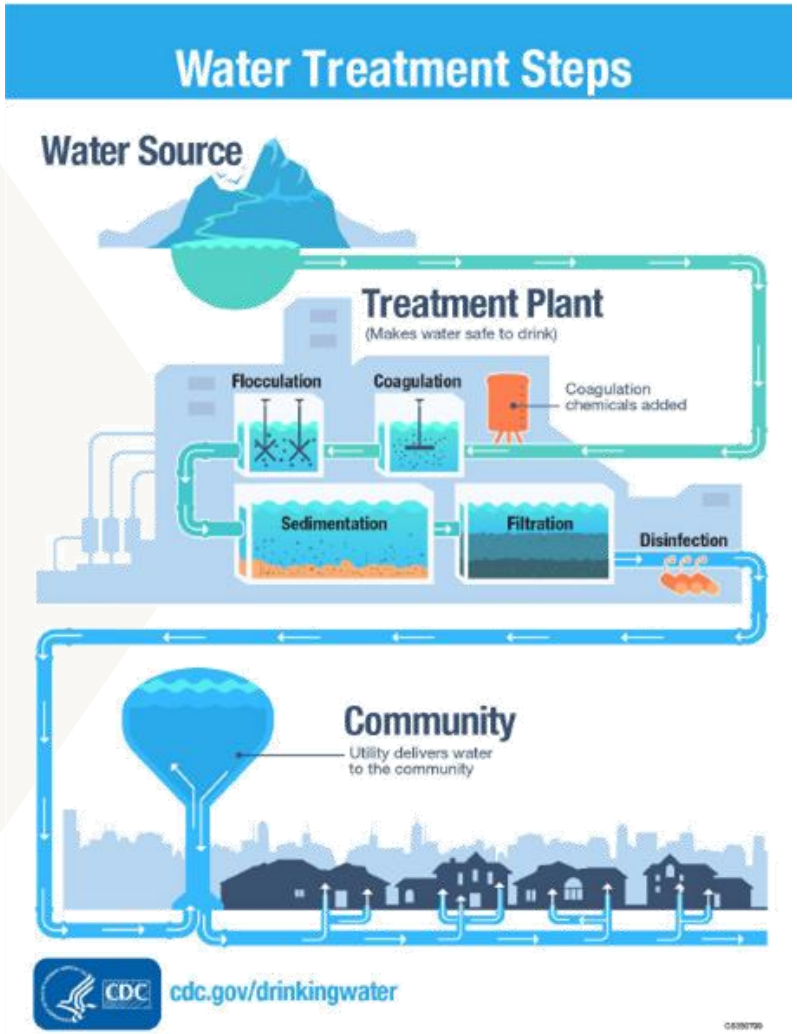
전통적 유해물질

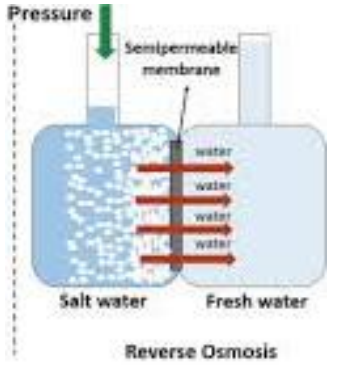
부유물, 영양분, 농약류, 내분비계 물질, 유기 화합물, 비소, 염

산업화에 따른 유해물질 (emerging contaminants)

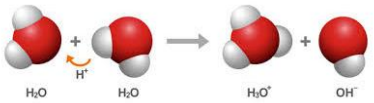
중금속, 과불화화합물

수처리 기술





역삼투
해수담수화



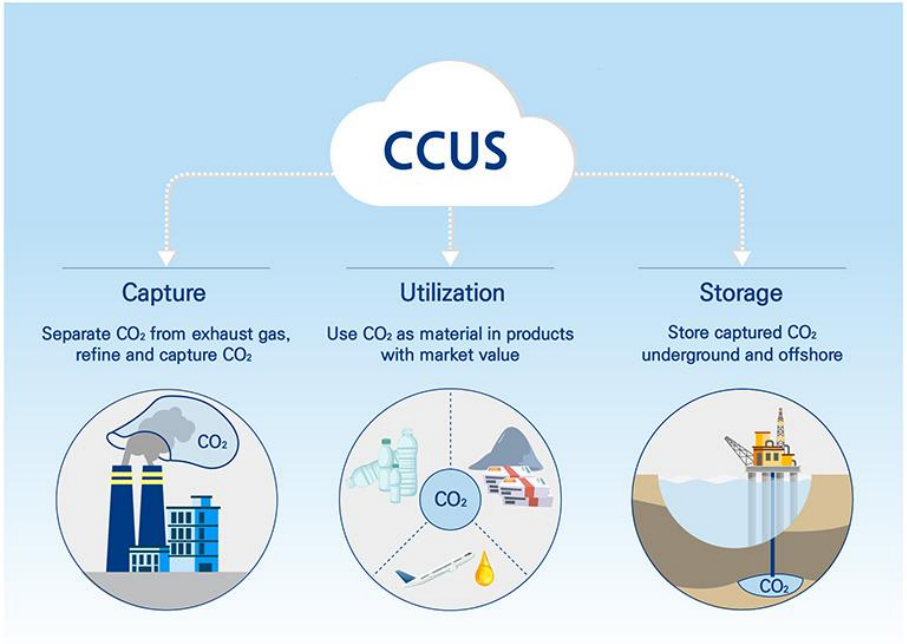
고도산화공정

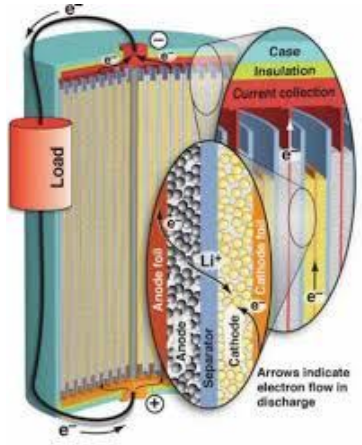
<출처: 미국 질병관리처>

환경-에너지 넥서스 기술 (탄소 중립, 지속 가능성 기술)



<출처: KISO 저널>

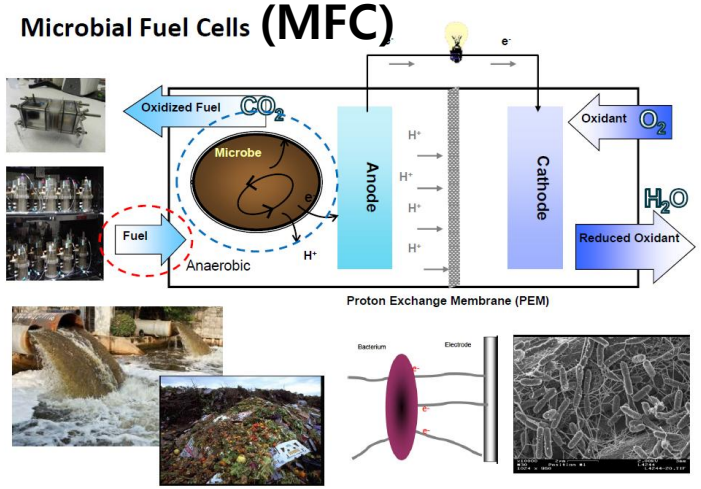


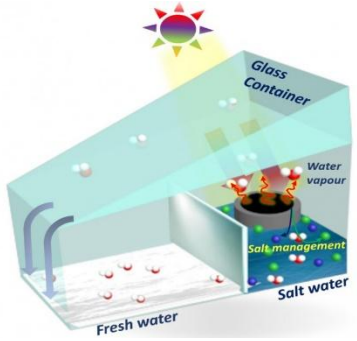


이차 전지

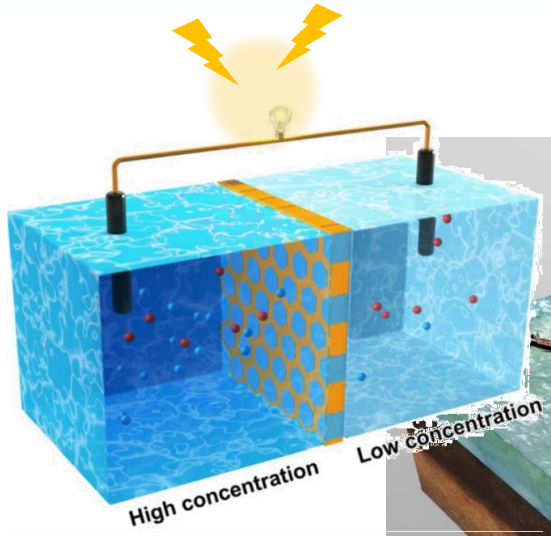
미생물 연료 전지

Microbial Fuel Cells (MFC)

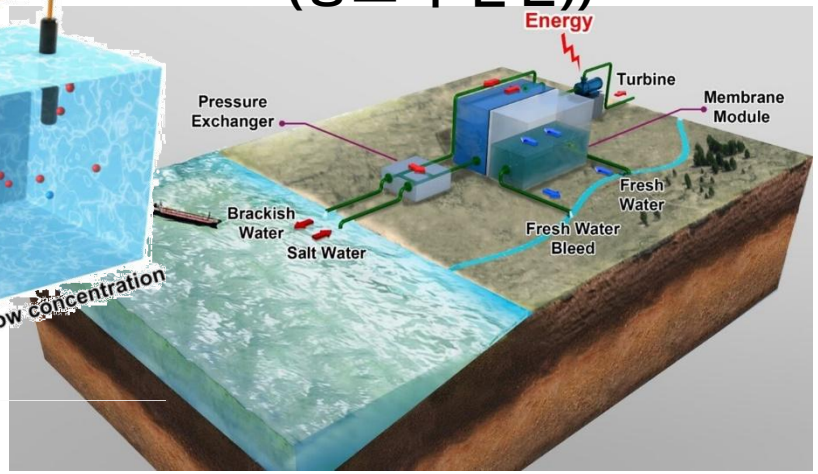




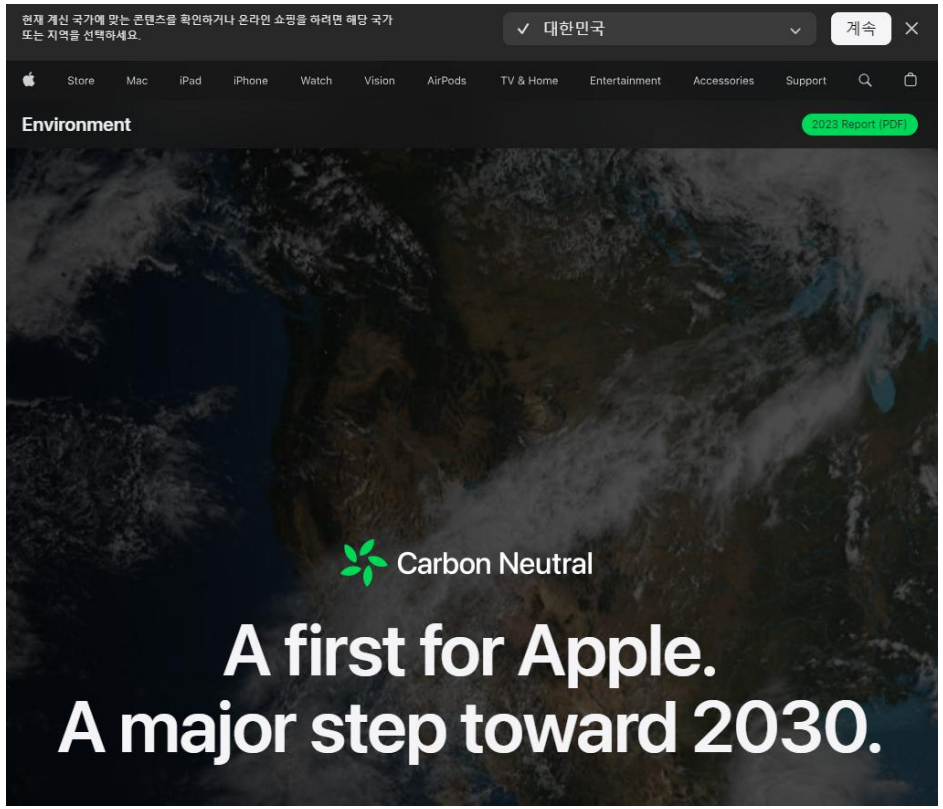
태양에너지
해수 담수화



블루 에너지 하베스팅 (농도차 발전))



Net Zero??



<출처: 애플>

Environment



Environment | Mother Nature

Learn more about Apple 2030 ›

https://www.apple.com/environment/mother-nature/

Net Zero??



https://www.apple.com/environment/mother-nature/

Topics in Environmental Engineering

- Provision of safe, palatable, and ample public water supplies
- Proper disposal of or recycling of wastewater and solid wastes
- Control of water, soil, and atmospheric pollution
  (including noise as an atmospheric pollutant)

Driving Forces in Modern Environmental Science & Engineering

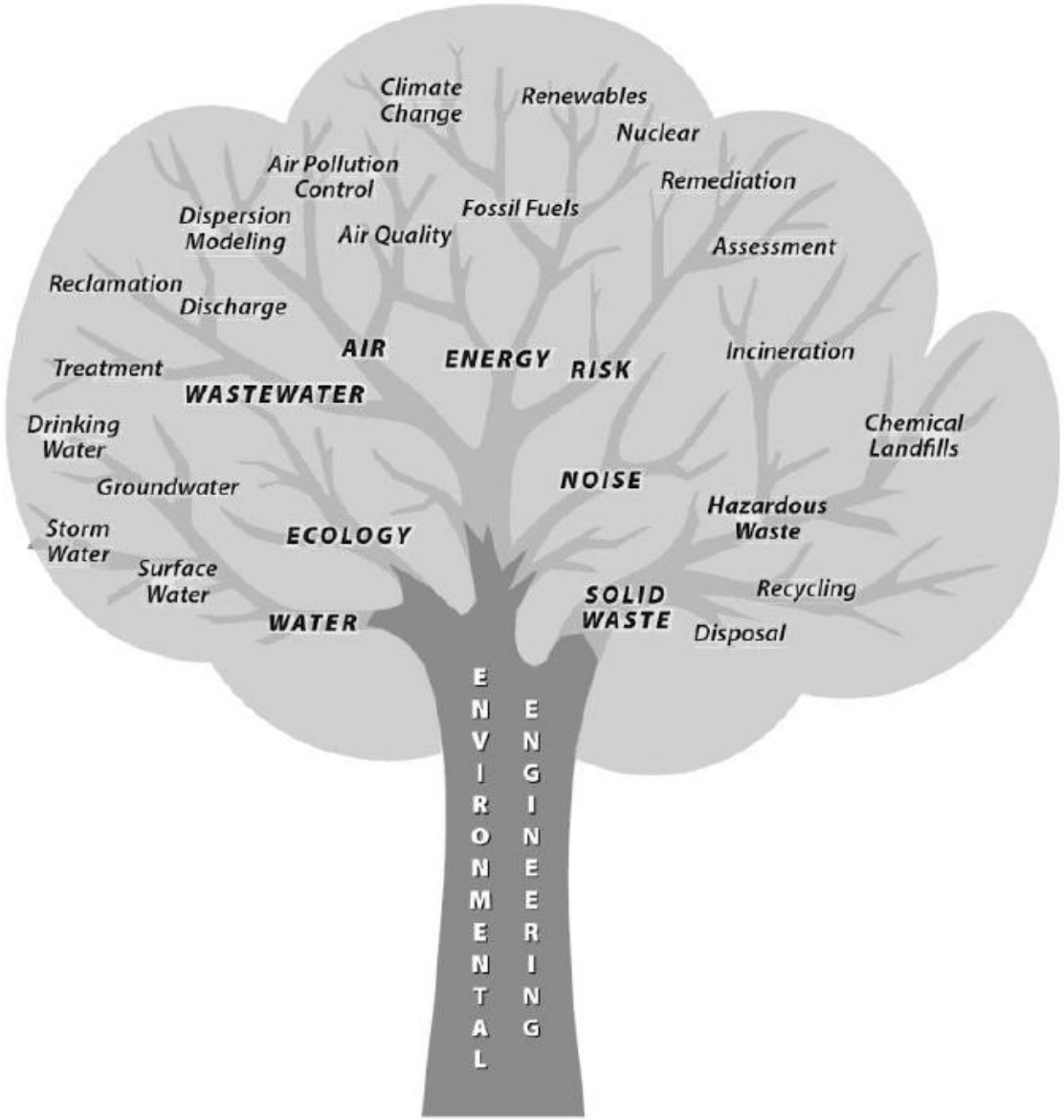
Human Health
- Sanitation
- Micropollutants

Social & Psychological Health
- Taste & Odor
- Envy Factor

Legal Aspect

Economical Aspect among others

환경과학 / 공학 Keywords



환경과학/공학의 '전통적' Keywords

- 정수, 하수, 폐기물, 대기오염, 생태,…

환경과학/공학의 '미래' Keywords

- AI, Big Data..
- 기후변화, 탄소중립
- 환경-에너지-자원 넥서스